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Studierichting: Informatica
Oefeningenreeks 5E nr. 2

$$\begin{aligned} S &= 4 \int_{\frac{\pi}{2}}^0 |a \sin^3 t| \cdot |-3a \cos^2 t \sin t| dt \\ &= -12a^2 \int_{\frac{\pi}{2}}^0 \sin^4 t \cos^2 t dt = 12a^2 \int_0^{\frac{\pi}{2}} \sin^4 t \cos^2 t dt \end{aligned}$$

$$\text{Gebruik: } \sin^2 t = \frac{1 - \cos 2t}{2} \text{ en } \cos^2 t = \frac{1 + \cos 2t}{2}$$

$$\Rightarrow \sin^4 t \cos^2 t = \left(\frac{1 - \cos 2t}{2} \right)^2 \cdot \left(\frac{1 + \cos 2t}{2} \right) = \frac{(1 - \cos 2t)^2 (1 + \cos 2t)}{8}$$

$$S = 12a^2 \cdot \frac{1}{8} \int_0^{\frac{\pi}{2}} (1 - \cos 2t)^2 (1 + \cos 2t) dt = \frac{3a^2}{2} \int_0^{\frac{\pi}{2}} (1 - \cos 2t)^2 (1 + \cos 2t) dt$$

Bereken de integrand:

$$(1 - \cos 2t)^2 (1 + \cos 2t) = (1 - 2 \cos 2t + \cos^2 2t)(1 + \cos 2t)$$

$$= 1 - \cos 2t - \cos^2 2t + \cos^3 2t$$

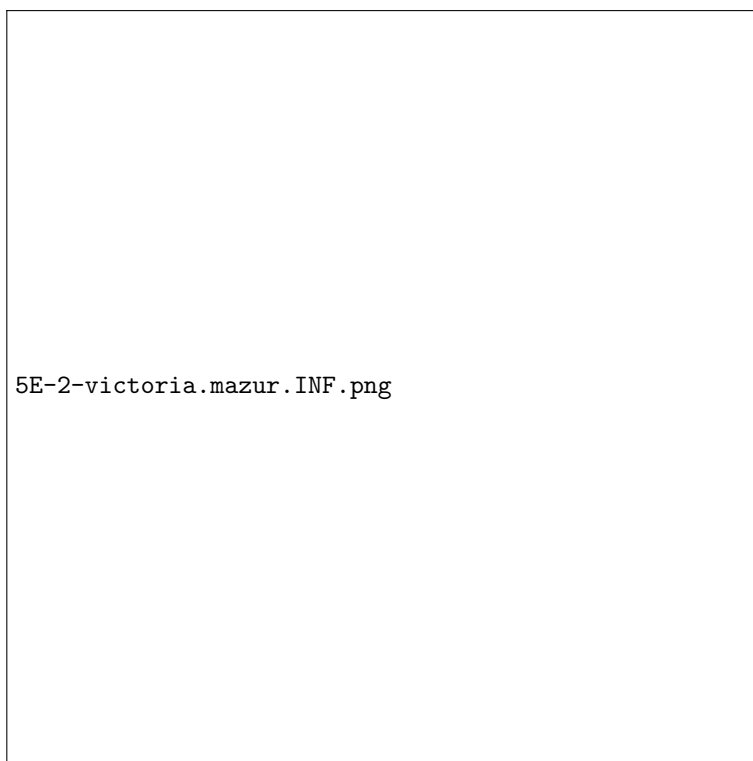
$$S = \frac{3a^2}{2} \int_0^{\frac{\pi}{2}} (1 - \cos 2t - \cos^2 2t + \cos^3 2t) dt$$

Gebruik bekende waarden:

$$\int_0^{\frac{\pi}{2}} \cos 2t dt = 0, \int_0^{\frac{\pi}{2}} \cos^2 2t dt = \frac{\pi}{4}, \int_0^{\frac{\pi}{2}} \cos^3 2t dt = 0$$

$$S = \frac{3a^2}{2} \left(\frac{\pi}{2} - 0 - \frac{\pi}{4} + 0 \right) = \frac{3a^2}{2} \cdot \frac{\pi}{4} = \frac{3\pi a^2}{8}$$

References



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